AI-Assisted Software Development

Module 3: Working with aider

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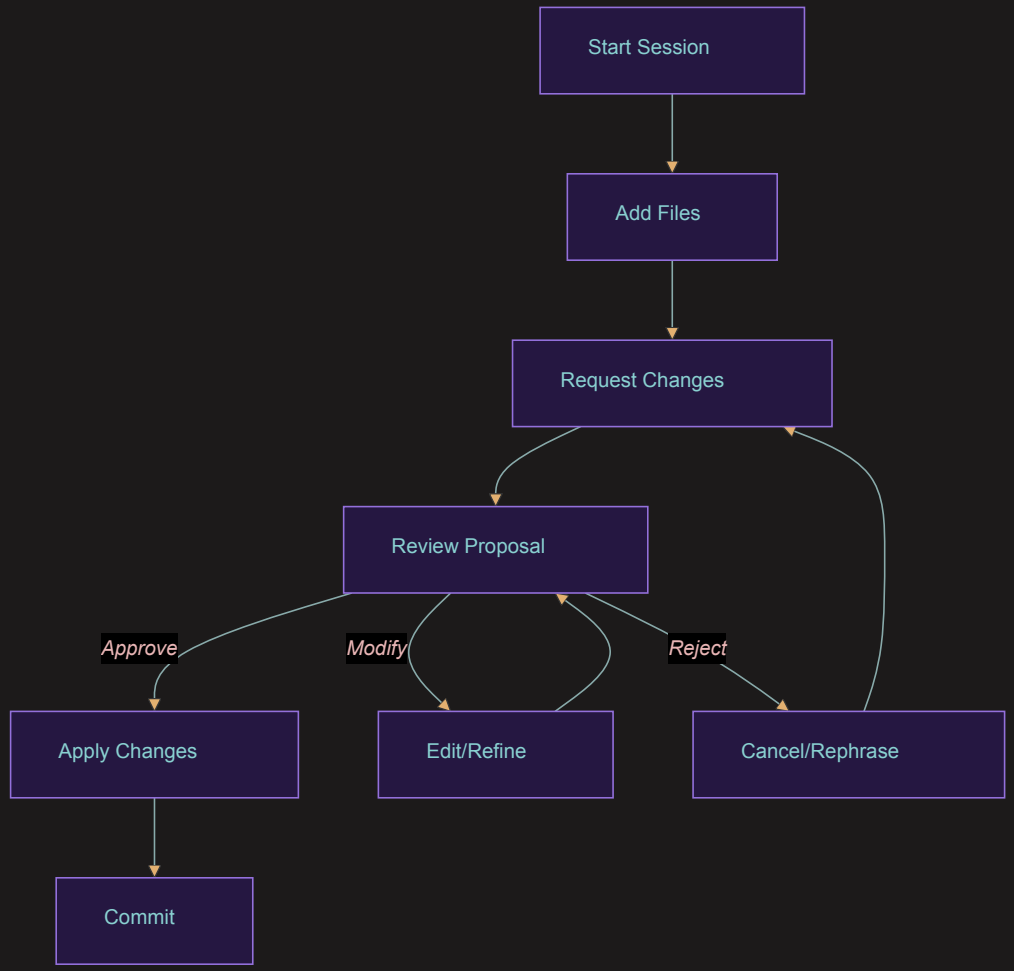
The aider Workflow



Effective Communication

DO:

- Be specific
- Provide context
- Break down complex tasks
- Reference file names
- Mention function names

DON'T:

- Be too vague
- Assume knowledge
- Request multiple unrelated changes
- Skip important details

Example:

```
> In user_auth.py, add password strength validation
  to the validate_password function. It should:
  - Require minimum 8 characters
  - Include at least one number
  - Include at least one special character
```

Working with Code Blocks

Different ways to reference code:

1. Direct Reference:

```
> Update the MAX_USERS constant in config.py to 100
```

2. Code Block Reference:

```
> Replace the sort_users function with this implementation:
def sort_users(users, key='name'):
    return sorted(users, key=lambda x: x[key])
```

3. Partial Updates:

```
> Add error handling to the process_data function
  around the file reading operation
```

Pattern Recognition

aider understands common coding patterns:

1. Function Modifications:

```
> Add input validation to process_user_data
```

2. Error Handling:

```
> Wrap the database operations in try/except blocks
```

3. Documentation:

```
> Add docstrings to all functions in auth.py
```

4. Refactoring:

```
> Extract the email validation logic into a separate function
```

Working with Multiple Files

Example: Updating related files

```
> Add a new user role system:
  1. Add Role enum in models.py
  2. Update User class to include role
  3. Modify authentication in auth.py
```

aider will:

1. Analyze dependencies
2. Make coordinated changes
3. Maintain consistency
4. Commit all changes together

Code Review Patterns

When reviewing aider's changes:

1. Check Logic:
    - Is the implementation correct?
    - Are edge cases handled?
    - Is error handling appropriate?
2. Check Style:
    - Does it match project style?
    - Is it well-documented?
    - Is it readable?
3. Check Integration:
    - Do changes work with existing code?
    - Are imports handled correctly?
    - Are dependencies updated?

Iterative Development

Example workflow:

1. Initial Request:

```
> Create a function to validate email addresses
```

2. Refine Implementation:

```
> Add support for custom domains
```

3. Add Features:

```
> Add blacklist checking for spam domains
```

4. Optimize:

```
> Cache the domain validation results
```

Working with Tests

aider can help with testing:

1. Creating Tests:

```
> Write unit tests for the email_validator function
```

2. Updating Tests:

```
> Update tests to cover the new domain blacklist feature
```

3. Test Fixes:

```
> Fix the failing test in test_email_validator.py
```

Error Handling

Common scenarios and solutions:

1. Unclear Response:

```
> Can you clarify what you mean by custom domains?
```

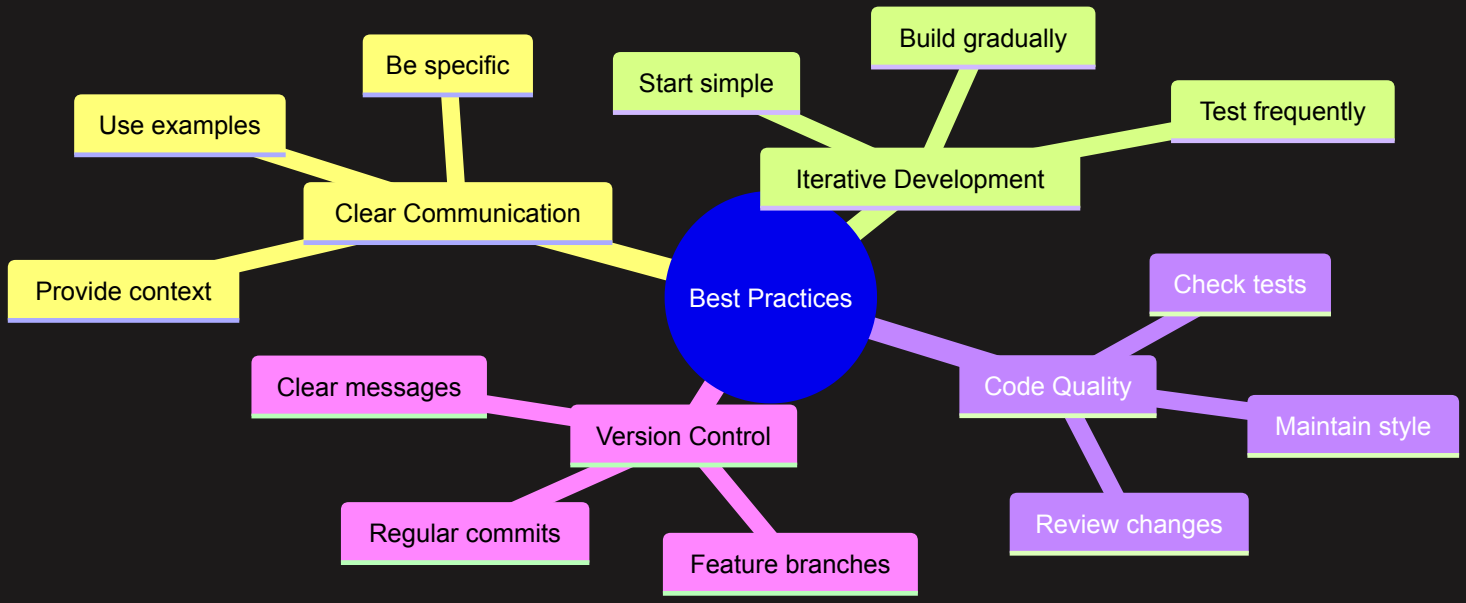
2. Incorrect Changes:

```
> That's not quite right. The validation should
  check for the @ symbol first
```

3. Partial Success:

```
> The domain validation looks good, but we also
  need to check for valid usernames
```

Best Practices



Common Patterns

1. Code Generation:

```
> Create a new class for handling API requests
```

2. Refactoring:

```
> Extract this logic into a separate utility function
```

3. Documentation:

```
> Add type hints and docstrings to this class
```

4. Bug Fixing:

```
> Fix the edge case where user_id is None
```

Advanced Communication

1. Providing Context:

```
> We're using SQLAlchemy for our ORM. Update the
  User model to include these new fields...
```

2. Referencing Standards:

```
> Following PEP 484, add type hints to all functions
```

3. Explaining Requirements:

```
> Due to GDPR requirements, we need to add data
  encryption to these user fields...
```

Recap: Working with aider

- Understanding core concepts
- Following effective workflows
- Communication patterns
- Working with code
- Handling multiple files
- Best practices
- Troubleshooting

Next Steps

- Exploring advanced features
- Custom configurations
- Integration with tools
- Complex refactoring

Questions to Consider:

- How can you improve your communication?
- What patterns work best for your projects?
- How can you optimize your workflow?